var, x, y, z
$label$, L, $Next$, $Done$
lit
f
$dsort$, D
$tvar$, a, b
$index$, i, j, k, n

terminals	::=		
	|	$\rightarrow$	
	|	$\rightsquigarrow$	
	|	$\rightarrow$	
	|	$\mid$	
	|	$\vee$	
	|	@	
	|	$\multimap$	
	|	$\wedge$	
	|	$\Rightarrow$	
	|	$\in$	
	|	true	
	|	false	
blit	::=		boolean literals
	|	true	
	|	false	
$A,\ B,\ C$	::=		types
	|	int	integers
	|	bool	booleans
	|	ptr A	pointer
	|	$(A_1, .., A_n)$	record
	|	$D(A_1, .., A_n)$	datatype application
	|	a	type variable
$\overline{A}$	::=		type lists
	|	$[]$	
	|	$A{::}\overline{A}$	
$\mathit{eff},\ \epsilon$	::=		effects
	|	**0**	
	|	**1**	
	|	**S**	
τ	::=		sorts
	|	int	integers
	|	bool	booleans
	|	ptr τ'	pointer
	|	$(\tau_1, .., \tau_n)$	tuple
	|	$D(\tau_1, .., \tau_n)$	datasort application
	|	pred τ'	predicate sort
	|	a	type variable
F	::=		function signatures
	|	$\tau \rightarrow_{\mathit{eff}} \tau'$	
Γ	::=		contexts
	|	$\cdot$	

	$\mid$	$\Gamma, x{:}_{\mathit{eff}}\tau$	
	$\mid$	Γ, Γ'	
Σ_c	$::=$		core signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_c, f : F$	
	$\mid$	$\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$	
	$\mid$	$\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	$\mid$	$\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Σ_s	$::=$		surface signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_s, f : F$	
	$\mid$	$\Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\}$	
	$\mid$	$\Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	$\mid$	$\Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Σ_r	$::=$		refined signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_r, f : F$	
	$\mid$	$\Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$	
	$\mid$	$\Sigma_r, f : RF$	
Φ	$::=$		type variable contexts
	$\mid$	$\cdot$	
	$\mid$	Φ, a	
	$\mid$	$\overline{a}$	
prim	$::=$		primitives
	$\mid$	$\mathbf{Add}$	addition
	$\mid$	$\mathbf{Sub}$	subtraction
	$\mid$	$\mathbf{Mul}$	multiplication
	$\mid$	$\mathbf{Div}$	division
	$\mid$	$\mathbf{And}$	conjunction
	$\mid$	$\mathbf{Or}$	disjunction
	$\mid$	$\mathbf{Not}$	negation
	$\mid$	$\mathbf{Lt}$	less than
	$\mid$	$\mathbf{Le}$	less or equal
	$\mid$	$\mathbf{Gt}$	greater than
	$\mid$	$\mathbf{Ge}$	greater or equal
	$\mid$	$\mathbf{Eq}\,[A]$	equality test
	$\mid$	$\mathbf{New}\,[A]$	allocate
	$\mid$	$\mathbf{Del}\,[A]$	deallocate
	$\mid$	$\mathbf{Get}\,[A]$	dereference
	$\mid$	$\mathbf{Set}\,[A]$	update
	$\mid$	$\mathbf{Own}\,[A]$	ownership
ce	$::=$		core expressions
	$\mid$	x	

	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, x = ce_1; ce_2$	
	$\mid$	$(ce_1, .., ce_n)$	
	$\mid$	$\mathbf{let}\, (x_1, .., x_n) = ce_1; ce_2$	
	$\mid$	$L\, ce$	
	$\mid$	$\mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 \mid .. \mid L_n\, x_n \to ce_n\}$	
	$\mid$	$\mathbf{iter}\, (x = ce_1)\{ce_2\}$	
	$\mid$	$prim\, ce$	
	$\mid$	$\mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3$	conditional
	$\mid$	$ce : \tau$	
	$\mid$	$f\, ce$	
	$\mid$	$ce_1 \doteq ce_2$	
	$\mid$	$\mathbf{take}\, x = ce_1; ce_2$	
	$\mid$	$\mathbf{return}\, ce$	
	$\mid$	(ce)	
	$\mid$	$[ce/x]ce'$	
se	$::=$		surface expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, pat = se_1; se_2$	
	$\mid$	$(se_1, .., se_n)$	
	$\mid$	$L\, se$	
	$\mid$	$\mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 \mid .. \mid pat_n \to se_n\}$	
	$\mid$	$\mathbf{iter}\, (pat = se_1)\{se_2\}$	
	$\mid$	$prim\, se$	
	$\mid$	$\mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3$	conditional
	$\mid$	$se : \tau$	
	$\mid$	$f\, se$	
	$\mid$	$se_1 \doteq se_2$	
	$\mid$	$\mathbf{take}\, pat = se_1; se_2$	
	$\mid$	$\mathbf{return}\, se$	
$prog$	$::=$		surface programs
	$\mid$	$\mathsf{fun}\, f : \tau \to_{eff} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} prog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} prog$	
	$\mid$	$\mathsf{main} : \tau_{eff} = se$	
$cprog$	$::=$		core programs
	$\mid$	$\mathsf{fun}\, f(x : \tau) \to_{eff} \tau' = ce \quad cprog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} cprog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} cprog$	
	$\mid$	$\mathsf{main} : \tau_{eff} = ce$	
pat	$::=$		patterns
	$\mid$	x	
	$\mid$	$L\, pat$	

	$\mid$	$(pat_1, .., pat_n)$	
β	$::=$		pattern bindings
	$\mid$	$\cdot$	
	$\mid$	$pat : \tau, \beta$	
	$\mid$	$pat_1 : \tau_1, .., pat_n : \tau_n, \beta$	
	$\mid$	$pat : \tau$	
$\mathcal{E}$	$::=$		evaluation contexts
	$\mid$	$[\cdot]$	
	$\mid$	$\mathbf{let}\, x = ce; \mathcal{E}$	
$\overline{y}$	$::=$		scrutinee variables
	$\mid$	$\cdot$	
	$\mid$	$y, \overline{y}$	
	$\mid$	$y_1, .., y_n, \overline{y}$	
M	$::=$		match branches
	$\mid$	$\cdot$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se \mid M$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se$	
$formula$	$::=$		
	$\mid$	$judgement$	
	$\mid$	$x{:}_{eff}\tau \in \Gamma$	
	$\mid$	$f{:}\tau \rightarrow_{eff} \tau' \in \Sigma_c$	
	$\mid$	A eqtype	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c$	
	$\mid$	$\tau = \mathbf{subst}\, \tau_1/a_1, .., \tau_i/a_i\, \mathbf{in}\, \tau'$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i)\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$L{:}A \in \mathsf{tyctors}(D) \in \Sigma_c$	
	$\mid$	$A = \mathbf{subst}\, A_1/a_1, .., A_i/a_i\, \mathbf{in}\, A'$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_k) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}\, \mathbf{in}\, \Sigma_c$	
	$\mid$	$\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c$	
	$\mid$	$a \in \Phi$	
	$\mid$	$L \neq L'$	
	$\mid$	$M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se}$	
	$\mid$	$ce = \mathcal{E}[ce']$	
	$\mid$	$D \neq D'$	
	$\mid$	$x \neq y$	
	$\mid$	$n = k$	
	$\mid$	$eff = eff_1 \vee eff_2$	
	$\mid$	$eff = eff_1 \vee eff_2 \vee eff_3$	
	$\mid$	$eff = eff_1 \vee .. \vee eff_n$	
	$\mid$	$\Gamma = x_1 : \tau_1[eff_1], .., x_n : \tau_n[eff_n]$	
	$\mid$	$formula_1 \quad .. \quad formula_n$	
	$\mid$	$x{:}ce\ [\mathsf{log}] \in \Delta$	
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}(u)] \in \Delta$	

	$\mid$	$x{:}_{\mathit{eff}}\tau \in \Delta$
	$\mid$	$u \in \{1, ?\}$
	$\mid$	$u \in \{0, ?\}$
	$\mid$	$u = u_1 \sqcap u_2$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Gamma = \mathsf{Bind}(\cdot; Pf)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_c$
	$\mid$	$\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\}$
	$\mid$	$f{:}RF \in \Sigma_c$
	$\mid$	$f{:}RF \in \Sigma_r$
	$\mid$	$f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_r$
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_r$
	$\mid$	$\mathsf{len}(\Delta) = \mathsf{len}(\Delta')$
	$\mid$	$A = \mathsf{Prod}(\overline{A})$
u	$::=$	usage flags
	$\mid$	0
	$\mid$	1
	$\mid$	$?$
Δ	$::=$	refined contexts
	$\mid$	$\cdot$
	$\mid$	$\Delta, x{:}_{\mathit{eff}}\tau$
	$\mid$	$\Delta, x{:}ce\ [\mathsf{log}]$
	$\mid$	$\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]$
	$\mid$	Δ, Δ'
$\overline{\Delta}$	$::=$	context lists
	$\mid$	$\Delta; \overline{\Delta}$
	$\mid$	$\cdot$
	$\mid$	$(\overline{\Delta})$
	$\mid$	$\Delta_1;\ ..\ ; \Delta_n$
$\mathcal{C}$	$::=$	SMT constraints
	$\mid$	$\top$
	$\mid$	$\mathcal{C}_1 \wedge \mathcal{C}_2$
	$\mid$	$\forall x{:}\tau.\, \mathcal{C}$
	$\mid$	$ce \Rightarrow \mathcal{C}$
	$\mid$	ce
	$\mid$	$\bot$
	$\mid$	$(\mathcal{C})$
Pf	$::=$	proof sorts
	$\mid$	$\cdot$
	$\mid$	$x{:}_{\mathit{eff}}\tau, Pf$
	$\mid$	$x{:}ce\ [\mathsf{log}], Pf$
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}], Pf$

	$\mid$	$[ce/x]Pf$	
RF	$::=$		refined function types
	$\mid$	$Pf_1 \multimap Pf_2\ [eff]$	
q	$::=$		refined patterns
	$\mid$	$()$	
	$\mid$	(x, q)	
crt	$::=$		core refined terms
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\ q = crt_1; crt_2$	
	$\mid$	$crt : Pf$	
	$\mid$	$prim\ rsp$	
	$\mid$	$f\ rsp$	
	$\mid$	(rsp)	
	$\mid$	$\mathbf{iter}\,[ce](q = crt_1)\{crt_2\}$	
	$\mid$	$\mathbf{if}\,[x]ce\,\mathbf{then}\,crt_1\,\mathbf{else}\,crt_2$	
	$\mid$	$\mathbf{case}\,[y]ce\,\mathbf{of}\,\{L_1\,x_1 \to crt_1 \mid .. \mid L_n\,x_n \to crt_n\}$	
	$\mid$	$\mathsf{exfalso}$	
	$\mid$	$\mathsf{open\text{-}take}\ rpf$	
rsp	$::=$		refined spines
	$\mid$	$\cdot$	
	$\mid$	ce, rsp	
	$\mid$	rpf, rsp	
	$\mid$	lpf, rsp	
lpf	$::=$		logical proof facts
	$\mid$	x	
	$\mid$	auto	
	$\mid$	$\mathsf{unfold}\ f(ce)$	
	$\mid$	$\mathsf{open\text{-}ret}\ rpf$	
	$\mid$	$lpf : ce$	
rpf	$::=$		resource proof facts
	$\mid$	x	
	$\mid$	$\mathsf{make\text{-}ret}\ lpf$	
	$\mid$	$\mathsf{make\text{-}take}\ crt$	
	$\mid$	$rpf : ce@ce'$	
$Jtype$	$::=$		
	$\mid$	$\Sigma_c; \Phi \vdash A\ \mathsf{wf}$	
	$\mid$	$\Sigma_c; \Phi; D' \vdash A\ \mathsf{guarded}$	
	$\mid$	$\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}$	
	$\mid$	$\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}$	
	$\mid$	$\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}$	

	$\mid$	A eqtype
	$\mid$	$\mathit{eff} \leq \mathit{eff}'$
	$\mid$	$\mathit{eff}' = \lfloor \mathit{eff} \rfloor$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	$\mid$	$\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Rightarrow \tau$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau$
	$\mid$	$\Sigma_c \vdash \mathit{cprog}$
Jcore	$::=$	
	$\mid$	$L : \tau \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c$
	$\mid$	$D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} \in \Sigma_c$
$\mathit{Jelaborate}$	$::=$	
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Rightarrow \tau \rightsquigarrow \mathit{ce}$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}$
	$\mid$	$\Sigma_s \vdash \mathit{prog}$ $\rightsquigarrow \Sigma_c \vdash \mathit{cprog}$
Jmatch	$::=$	
	$\mid$	M has_con
	$\mid$	M has_tup
	$\mid$	$M/\mathit{var} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathit{ce}$
	$\mid$	$\mathcal{E}[\mathit{ce}] \rightsquigarrow \mathit{ce}'$
$\mathit{Jrefine}$	$::=$	
	$\mid$	$\Sigma_c \vdash \Delta$ wf
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$\Sigma_c; \Gamma \vdash \mathcal{C}$ wf
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}$ wf
	$\mid$	$\Sigma_c \vdash \mathit{RF}$ wf
	$\mid$	$\Sigma_r \vdash f{:}\mathit{RF}$
	$\mid$	$\overline{A} = \{\mathit{Pf}\}$
	$\mid$	$\Sigma_c = \mathsf{Comp}(\Sigma_r)$
	$\mid$	$\vdash \Sigma_r$ wf
	$\mid$	$\Sigma_c; \Delta \vdash \mathit{lpf} \Rightarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash x : \mathit{ce}@\mathit{ce}' \dashv \Delta'$
	$\mid$	$\Sigma_c; \Delta \vdash \mathit{rpf} \Rightarrow \mathit{ce}@\mathit{ce}' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash \mathit{rpf} \Leftarrow \mathit{ce}@\mathit{ce}' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash \mathit{Pf}_1 = \mathit{Pf}_2 \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta$
	$\mid$	$\Sigma_c; \Gamma \vdash q : \mathit{Pf} \dashv \Delta$
	$\mid$	$\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash_{\mathit{eff}} y{:}\mathit{ce}$ is $L\,x \rightarrow \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}$

	$\mid$	$\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{RF} \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_c \vdash \sqcap\overline{\Delta} = \Delta$
judgement	$::=$	
	$\mid$	*Jtype*
	$\mid$	*Jcore*
	$\mid$	*Jelaborate*
	$\mid$	*Jmatch*
	$\mid$	*Jrefine*
user_syntax	$::=$	
	$\mid$	*var*
	$\mid$	*label*
	$\mid$	*lit*
	$\mid$	*f*
	$\mid$	*dsort*
	$\mid$	*tvar*
	$\mid$	*index*
	$\mid$	*terminals*
	$\mid$	*blit*
	$\mid$	A
	$\mid$	$\overline{A}$
	$\mid$	*eff*
	$\mid$	τ
	$\mid$	F
	$\mid$	Γ
	$\mid$	Σ_c
	$\mid$	Σ_s
	$\mid$	Σ_r
	$\mid$	Φ
	$\mid$	*prim*
	$\mid$	*ce*
	$\mid$	*se*
	$\mid$	*prog*
	$\mid$	*cprog*
	$\mid$	*pat*
	$\mid$	β
	$\mid$	$\mathcal{E}$
	$\mid$	$\overline{y}$
	$\mid$	M
	$\mid$	*formula*
	$\mid$	u
	$\mid$	Δ
	$\mid$	$\overline{\Delta}$
	$\mid$	$\mathcal{C}$
	$\mid$	Pf
	$\mid$	RF
	$\mid$	q

$$\begin{array}{ll} | & crt \\ | & rsp \\ | & lpf \\ | & rpf \end{array}$$

$\boxed{\Sigma_c; \Phi \vdash A \text{ wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \text{int wf}} \quad \textsc{TWf_int}$$

$$\frac{}{\Sigma_c; \Phi \vdash \text{bool wf}} \quad \textsc{TWf_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A \text{ wf}}{\Sigma_c; \Phi \vdash \text{ptr } A \text{ wf}} \quad \textsc{TWf_ptr}$$

$$\frac{\Sigma_c; \Phi \vdash A_1 \text{ wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \text{ wf}}{\Sigma_c; \Phi \vdash (A_1, .., A_n) \text{ wf}} \quad \textsc{TWf_record}$$

$$\frac{\begin{array}{c}\textbf{type}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash A_1 \text{ wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \text{ wf}\end{array}}{\Sigma_c; \Phi \vdash D(A_1, .., A_n) \text{ wf}} \quad \textsc{TWf_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a \text{ wf}} \quad \textsc{TWf_tvar}$$

$\boxed{\Sigma_c; \Phi; D' \vdash A \text{ guarded}}$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \text{int guarded}} \quad \textsc{TWfG_int}$$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \text{bool guarded}} \quad \textsc{TWfG_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A \text{ wf}}{\Sigma_c; \Phi; D' \vdash \text{ptr } A \text{ guarded}} \quad \textsc{TWfG_ptr}$$

$$\frac{\Sigma_c; \Phi; D' \vdash A_1 \text{ guarded} \quad .. \quad \Sigma_c; \Phi; D' \vdash A_n \text{ guarded}}{\Sigma_c; \Phi; D' \vdash (A_1, .., A_n) \text{ guarded}} \quad \textsc{TWfG_record}$$

$$\frac{\begin{array}{c}\textbf{type}\, D(a_1, .., a_k)\, \textbf{in}\, \Sigma_c \quad n = k \quad D \neq D' \\ \Sigma_c; \Phi \vdash A_1 \text{ wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \text{ wf}\end{array}}{\Sigma_c; \Phi; D' \vdash D(A_1, .., A_n) \text{ guarded}} \quad \textsc{TWfG_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi; D' \vdash a \text{ guarded}} \quad \textsc{TWfG_tvar}$$

$\boxed{\Sigma_c \vdash \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} \text{ ok}}$

$$\frac{\Sigma_c, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_1 \text{ wf} \quad .. \quad \Sigma_c, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_n \text{ wf}}{\Sigma_c \vdash \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} \text{ ok}} \quad \textsc{SDWf_ok}$$

$\boxed{\Sigma_c \vdash \text{type } D(\overline{a}) = \{\overline{L : A}\} \text{ ok}}$

$$\frac{\Sigma_c, \text{type } D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_1 \text{ guarded} \quad .. \quad \Sigma_c, \text{type } D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_n \text{ guarded}}{\Sigma_c \vdash \text{type } D(\overline{a}) = \{\overline{L : A}\} \text{ ok}} \quad \textsc{TDWf_ok}$$

$\boxed{\Sigma_c; \Phi \vdash \tau \text{ wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{int\ wf}} \quad \textsc{SWf_int}$$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{bool\ wf}} \quad \textsc{SWf_bool}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{ptr}\ \tau\ \mathsf{wf}} \quad \textsc{SWf_ptr}$$

$$\frac{\Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}}{\Sigma_c; \Phi \vdash (\tau_1, \ .., \tau_n)\ \mathsf{wf}} \quad \textsc{SWf_record}$$

$$\frac{\begin{array}{c}\mathbf{sort}\ D(a_1, \ .., a_k)\ \mathbf{in}\ \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}\end{array}}{\Sigma_c; \Phi \vdash D(\tau_1, \ .., \tau_n)\ \mathsf{wf}} \quad \textsc{SWf_app}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{pred}\ \tau\ \mathsf{wf}} \quad \textsc{SWf_pred}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a\ \mathsf{wf}} \quad \textsc{SWf_tvar}$$

$\boxed{A\ \mathsf{eqtype}}$

$$\frac{}{\mathsf{int\ eqtype}} \quad \textsc{EqTy_int}$$

$$\frac{}{\mathsf{bool\ eqtype}} \quad \textsc{EqTy_bool}$$

$$\frac{}{\mathsf{ptr}\ A\ \mathsf{eqtype}} \quad \textsc{EqTy_ptr}$$

$\boxed{\mathit{eff} \leq \mathit{eff}'}$

$$\frac{}{\mathbf{0} \leq \mathbf{0}} \quad \textsc{Sub_pure_pure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{1}} \quad \textsc{Sub_pure_impure}$$

$$\frac{}{\mathbf{1} \leq \mathbf{1}} \quad \textsc{Sub_impure_impure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{S}} \quad \textsc{Sub_pure_spec}$$

$$\frac{}{\mathbf{S} \leq \mathbf{S}} \quad \textsc{Sub_spec_spec}$$

$\boxed{\mathit{eff}' = \lfloor \mathit{eff} \rfloor}$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor} \quad \textsc{Pur_pure}$$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor} \quad \textsc{Pur_impure}$$

$$\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor} \quad \textsc{Pur_spec}$$

$\boxed{\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2}$

$$\frac{}{\mathit{eff} = \mathit{eff} \vee \mathit{eff}} \quad \textsc{Join_refl}$$

$$\frac{}{\mathbf{1} = \mathbf{0} \vee \mathbf{1}} \quad \textsc{Join_pure_impure}$$

$$\frac{}{\mathbf{1} = \mathbf{1} \vee \mathbf{0}} \quad \textsc{Join_impure_pure}$$

$$\frac{}{\mathbf{S} = \mathbf{0} \vee \mathbf{S}} \quad \textsc{Join_pure_spec}$$

$$\frac{}{\mathbf{S} = \mathbf{S} \vee \mathbf{0}} \quad \textsc{Join_spec_pure}$$

$\boxed{prim{:}\tau \rightarrow_{\mathit{eff}} \tau'}$

$$\frac{}{\mathbf{Add}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_add}$$

$$\frac{}{\mathbf{Sub}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_sub}$$

$$\frac{}{\mathbf{Mul}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}} \quad \textsc{PT_mul}$$

$$\frac{}{\mathbf{Div}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{1}} \mathsf{int}} \quad \textsc{PT_div}$$

$$\frac{}{\mathbf{And}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_and}$$

$$\frac{}{\mathbf{Or}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_or}$$

$$\frac{}{\mathbf{Not}{:}\mathsf{bool} \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_not}$$

$$\frac{}{\mathbf{Lt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_lt}$$

$$\frac{}{\mathbf{Le}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_le}$$

$$\frac{}{\mathbf{Gt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_gt}$$

$$\frac{}{\mathbf{Ge}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_ge}$$

$$\frac{A\ \mathsf{eqtype}}{\mathbf{Eq}\,[A]{:}(A, A) \rightarrow_{\mathbf{0}} \mathsf{bool}} \quad \textsc{PT_eq}$$

$$\frac{}{\mathbf{New}\,[A]{:}A \rightarrow_{\mathbf{1}} \mathsf{ptr}\ A} \quad \textsc{PT_new}$$

$$\frac{}{\mathbf{Del}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} (\,)} \quad \textsc{PT_del}$$

$$\frac{}{\mathbf{Get}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} A} \quad \textsc{PT_get}$$

$$\frac{}{\mathbf{Set}\,[A]{:}(\mathsf{ptr}\ A, A) \rightarrow_{\mathbf{1}} (\,)} \quad \textsc{PT_set}$$

$$\frac{}{\mathbf{Own}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{S}} \mathsf{pred}\ (A)} \quad \textsc{PT_own}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int}} \quad \textsc{Syn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool}} \quad \textsc{Syn_bool}$$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \qquad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau} \quad \textsc{Syn_var}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce : \tau \Rightarrow \tau} \quad \textsc{Syn_annot}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ ce \Rightarrow \tau'} \quad \textsc{Syn_app}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ ce \Rightarrow \tau'} \quad \textsc{Syn_call}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_1 \Rightarrow A \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_2 \Leftarrow A \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \doteq ce_2 \Rightarrow \mathsf{bool}} \quad \textsc{Syn_eq}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \Rightarrow \tau \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma, x{:}_\epsilon \tau \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\ x = ce_1; ce_2 \Leftarrow \tau'} \quad \textsc{Chk_let}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (ce_1, .., ce_n) \Leftarrow (\tau_1, .., \tau_n)} \quad \textsc{Chk_tuple}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_\epsilon ce_1 \Rightarrow (\tau_1, .., \tau_n) \\ \Gamma' = x_1 : \tau_1[\epsilon], .., x_n : \tau_n[\epsilon] \\ \Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\ (x_1, .., x_n) = ce_1; ce_2 \Leftarrow \tau'} \quad \textsc{Chk_lettuple}$$

$$\frac{\begin{array}{l} L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\ ce \Leftarrow D(\tau_1, .., \tau_i)} \quad \textsc{Chk_inject}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, .., \tau_i) \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1''\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \quad .. \quad L_n : \tau_n''\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ \Sigma_c; \Gamma, x_1{:}_{\mathit{eff}'} \tau_1'' \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau \quad .. \quad \Sigma_c; \Gamma, x_n{:}_{\mathit{eff}'} \tau_n'' \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\ ce\ \mathbf{of}\ \{L_1\ x_1 \rightarrow ce_1 | .. | L_n\ x_n \rightarrow ce_n\} \Leftarrow \tau} \quad \textsc{Chk_case}$$

$$\frac{\begin{array}{l} \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} ce_1 \Rightarrow A \\ \Sigma_c; \Gamma, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce_2 \Leftarrow D(A, B) \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\ (x = ce_1)\{ce_2\} \Leftarrow B} \quad \textsc{Chk_iter}$$

$$\frac{\begin{array}{c} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_1 \Leftarrow \mathsf{bool} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_3 \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\ ce_1\ \mathbf{then}\ ce_2\ \mathbf{else}\ ce_3 \Leftarrow \tau} \quad \textsc{Chk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\ ce \Leftarrow \mathsf{pred}\ \tau} \quad \textsc{Chk_return}$$

$$\frac{\begin{array}{c} \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{pred}\ \tau \\ \Sigma_c; \Gamma, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce_2 \Leftarrow \mathsf{pred}\ \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\ x = ce_1; ce_2 \Leftarrow \mathsf{pred}\ \tau'} \quad \textsc{Chk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau} \quad \textsc{Chk_sub}$$

$\boxed{\Sigma_c \vdash \mathit{cprog}}$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Leftarrow \tau}{\Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce} \quad \textsc{CProg_main}$$

$$\frac{\begin{array}{c} \Sigma_c; \Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_c, \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_purefun}$$

$$\frac{\begin{array}{c} \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_c, f : A \rightarrow_{\mathbf{1}} B \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{1}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_impurefun}$$

$$\frac{\begin{array}{c} \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \Gamma, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \\ \Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad \mathit{cprog}} \quad \textsc{CProg_specfun}$$

$$\frac{\begin{array}{c} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathbf{sort}\ D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{cprog}} \quad \textsc{CProg_sortdecl}$$

$$\frac{\begin{array}{c} \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathbf{type}\ D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{cprog}} \quad \textsc{CProg_typedecl}$$

$\boxed{L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c}$

$$\frac{\mathbf{sort}\ D(a_1, .., a_i)\ \mathbf{in}\ \Sigma_c \quad L{:}\tau' \in \mathrm{ctors}(D) \in \Sigma_c \quad \tau = \mathbf{subst}\ \tau_1/a_1, .., \tau_i/a_i\ \mathbf{in}\ \tau'}{L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c} \quad \textsc{Ctor_sort}$$

$$\frac{\mathbf{type}\ D(a_1, .., a_i)\ \mathbf{in}\ \Sigma_c \quad L{:}A' \in \mathsf{tyctors}(D) \in \Sigma_c \quad A = \mathbf{subst}\ A_1/a_1, .., A_i/a_i\ \mathbf{in}\ A'}{L : A\ \mathbf{in}\ D(A_1, .., A_i)\ \mathbf{in}\ \Sigma_c} \quad \textsc{Ctor_type}$$

$\boxed{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c}$

$$\frac{\mathbf{sort}\ D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\ \mathbf{in}\ \Sigma_c}{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c} \quad \textsc{DL_sort}$$

$$\frac{\mathbf{type}\ D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\}\ \mathbf{in}\ \Sigma_c}{D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\} \in \Sigma_c} \quad \textsc{DL_type}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau \rightsquigarrow x} \quad \textsc{SurfSyn_var}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int} \rightsquigarrow \mathit{lit}} \quad \textsc{SurfSyn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool} \rightsquigarrow \mathit{blit}} \quad \textsc{SurfSyn_bool}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} : \tau \Rightarrow \tau \rightsquigarrow \mathit{ce} : \tau} \quad \textsc{SurfSyn_annot}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce} \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ \mathit{se} \Rightarrow \tau' \rightsquigarrow \mathit{prim}\ \mathit{ce}} \quad \textsc{SurfSyn_app}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce} \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ \mathit{se} \Rightarrow \tau' \rightsquigarrow f\ \mathit{ce}} \quad \textsc{SurfSyn_call}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{se}_1 \Rightarrow A \rightsquigarrow \mathit{ce}_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{se}_2 \Leftarrow A \rightsquigarrow \mathit{ce}_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_1 \doteq \mathit{se}_2 \Rightarrow \mathsf{bool} \rightsquigarrow \mathit{ce}_1 \doteq \mathit{ce}_2} \quad \textsc{SurfSyn_eq}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_1 \Rightarrow \tau \rightsquigarrow \mathit{ce}_1 \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; \mathit{pat} : \tau \rightarrow [\cdot][[\cdot]]\mathit{se}_2 \Leftarrow_\epsilon \tau' \rightsquigarrow \mathit{ce}_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{let}\ \mathit{pat} = \mathit{se}_1; \mathit{se}_2 \Leftarrow \tau' \rightsquigarrow \mathbf{let}\ y = \mathit{ce}_1; \mathit{ce}_2} \quad \textsc{SurfChk_let}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_1 \Leftarrow \tau_1 \rightsquigarrow \mathit{ce}_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_n \Leftarrow \tau_n \rightsquigarrow \mathit{ce}_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (\mathit{se}_1, .., \mathit{se}_n) \Leftarrow (\tau_1, .., \tau_n) \rightsquigarrow (\mathit{ce}_1, .., \mathit{ce}_n)} \quad \textsc{SurfChk_tuple}$$

$$\frac{\begin{array}{l} L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce} \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\ \mathit{se} \Leftarrow D(\tau_1, .., \tau_i) \rightsquigarrow L\ \mathit{ce}} \quad \textsc{SurfChk_inject}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_\epsilon \mathit{se} \Rightarrow \tau \rightsquigarrow \mathit{ce}' \\ M = \overline{\mathit{pat}{:}\tau, \cdot \rightarrow [\cdot][\cdot]\,\mathit{se}} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; M \Leftarrow_\epsilon \tau' \rightsquigarrow \mathit{ce} \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\ \mathit{se}\ \mathbf{of}\ \{\mathit{pat}_1 \rightarrow \mathit{se}_1 | .. | \mathit{pat}_n \rightarrow \mathit{se}_n\} \Leftarrow \tau' \rightsquigarrow \mathbf{let}\ y = \mathit{ce}'; \mathit{ce}} \quad \textsc{SurfChk_case}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{0}} \mathit{se}_1 \Rightarrow A \rightsquigarrow \mathit{ce}_1 \\ \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{1}} z, \cdot; \mathit{pat} : A, \cdot \rightarrow [\cdot][[\cdot]]\mathit{se}_2 \Leftarrow_{\mathbf{0}} D(A, B) \rightsquigarrow \mathit{ce}_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\,(\mathit{pat} = \mathit{se}_1)\{\mathit{se}_2\} \Leftarrow B \rightsquigarrow \mathbf{iter}\,(z = \mathit{ce}_1)\{\mathit{ce}_2\}} \quad \textsc{SurfChk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Leftarrow \mathsf{bool} \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow \tau \rightsquigarrow ce_2 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_3 \Leftarrow \tau \rightsquigarrow ce_3 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3 \Leftarrow \tau \rightsquigarrow \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3} \quad \textsc{SurfChk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\, se \Leftarrow \mathsf{pred}\, \tau \rightsquigarrow \mathbf{return}\, ce} \quad \textsc{SurfChk_return}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{S}} se_1 \Rightarrow \mathsf{pred}\, \tau \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathbf{S}} y, \cdot; pat : \tau \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_{\mathbf{S}} \mathsf{pred}\, \tau' \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\, pat = se_1; se_2 \Leftarrow \mathsf{pred}\, \tau' \rightsquigarrow \mathbf{take}\, y = ce_1; ce_2} \quad \textsc{SurfChk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce} \quad \textsc{SurfChk_sub}$$

$$\boxed{\begin{array}{l} \Sigma_s \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathit{cprog} \end{array}}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}} se \Leftarrow \tau \rightsquigarrow ce}{\begin{array}{l} \Sigma_s \vdash \mathsf{main} : \tau_{\mathit{eff}} = se \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce \end{array}} \quad \textsc{Prog_main}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\, f : A \rightarrow_{\mathbf{0}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\, f(y : A) \rightarrow_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\, f : A \rightarrow_{\mathbf{0}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\, f(y : A) \rightarrow_{\mathbf{0}} B = ce \quad \mathit{cprog} \end{array}} \quad \textsc{Prog_purefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \Gamma \vdash_{\mathbf{1}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, f : A \rightarrow_{\mathbf{1}} B \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, f : A \rightarrow_{\mathbf{1}} B \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\, f : A \rightarrow_{\mathbf{1}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\, f(y : A) \rightarrow_{\mathbf{1}} B = ce \quad \mathit{cprog} \end{array}} \quad \textsc{Prog_impurefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \Gamma \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau' \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\, f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\, f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\, f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\, f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad \mathit{cprog} \end{array}} \quad \textsc{Prog_specfun}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{sort}\, D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{sort}\, D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, \mathsf{sort}\, D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{cprog} \end{array}} \quad \textsc{Prog_sortdecl}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{type}\, D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{type}\, D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, \mathsf{type}\, D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{cprog} \end{array}} \quad \textsc{Prog_typedecl}$$

$\boxed{M \ \mathsf{has_con}}$

$$\frac{}{L\,pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M \ \mathsf{has_con}} \quad \text{HC_HERE}$$

$$\frac{M \ \mathsf{has_con}}{\beta \to [\Gamma][\mathcal{E}]se \mid M \ \mathsf{has_con}} \quad \text{HC_THERE}$$

$\boxed{M \ \mathsf{has_tup}}$

$$\frac{}{(pat_1, \ldots, pat_n) : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M \ \mathsf{has_tup}} \quad \text{HT_HERE}$$

$$\frac{M \ \mathsf{has_tup}}{\beta \to [\Gamma][\mathcal{E}]se \mid M \ \mathsf{has_tup}} \quad \text{HT_THERE}$$

$\boxed{M/var \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/var \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{SV_EMPTY}$$

$$\frac{M/var \overset{y}{\rightsquigarrow}_{eff} M'}{x : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M/var \overset{y}{\rightsquigarrow}_{eff} \beta \to [\Gamma, x{:}_{eff}\tau][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SV_CONS}$$

$\boxed{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{SC_EMPTY}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{L\,pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{SC_MATCH}$$

$$\frac{L \neq L' \quad M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{L'\,pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'} \quad \text{SC_SKIP}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} M'}{x : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{eff} z : \tau, \beta \to [\Gamma, x{:}_{eff}\tau'][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SC_VAR}$$

$\boxed{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}$

$$\frac{}{\cdot/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} \cdot} \quad \text{ET_EMPTY}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}{(pat_1, \ldots, pat_k) : (\tau_1, \ldots, \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} pat_1 : \tau_1, \ldots, pat_k : \tau_k, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} M'}{x : (\tau_1, \ldots, \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{eff} z_1 : \tau_1, \ldots, z_k : \tau_k, \beta \to [\Gamma, x{:}_{eff}(\tau_1, \ldots, \tau_k)][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VA}$$

$\boxed{\Sigma_c; \Gamma \vdash_{eff_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce}$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{eff_0} se \Leftarrow \tau \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{eff_0} \cdot; \cdot \to [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_DONE}$$

$$\frac{\begin{array}{l} M/var \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{eff_0} \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{eff_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_VAR}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_con} \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \quad .. \quad L_n : \tau_n'' \ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ M/L_1{:}\tau_1'' \overset{y}{\rightsquigarrow}_\epsilon M_1 \quad .. \quad M/L_n{:}\tau_n'' \overset{y}{\rightsquigarrow}_\epsilon M_n \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_1, \overline{y}; M_1 \Leftarrow_\epsilon \tau \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_n, \overline{y}; M_n \Leftarrow_\epsilon \tau \rightsquigarrow ce_n \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{case}\ y\ \mathbf{of}\ \{L_1\ x_1 \to ce_1 | .. | L_n\ x_n \to ce_n\}} \quad \textsc{Cov_con}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_tup} \\ M/\mathbf{tup} \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y_1, .., y_n, \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{let}\ (y_1, .., y_n) = y; ce} \quad \textsc{Cov_tup}$$

$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \quad \textsc{Fill_hole}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\mathbf{let}\ x = ce''; \mathcal{E}[ce] \rightsquigarrow \mathbf{let}\ x = ce''; ce'} \quad \textsc{Fill_let}$$

$\boxed{\Sigma_c \vdash \Delta\ \mathsf{wf}}$

$$\frac{}{\Sigma_c \vdash \cdot\ \mathsf{wf}} \quad \textsc{RWf_empty}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf} \quad \mathit{eff} \leq \mathbf{S}}{\Sigma_c \vdash \Delta, x{:}_{\mathit{eff}}\tau\ \mathsf{wf}} \quad \textsc{RWf_eff}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool}}{\Sigma_c \vdash \Delta, x{:}ce\ [\mathsf{log}]\ \mathsf{wf}} \quad \textsc{RWf_log}$$

$$\frac{\Sigma_c \vdash \Delta\ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Leftarrow \tau}{\Sigma_c \vdash \Delta, x{:}ce@ce'\ [\mathsf{res}(u)]\ \mathsf{wf}} \quad \textsc{RWf_res}$$

$\boxed{\Gamma = |\Delta|}$

$$\frac{}{\cdot = |\cdot|} \quad \textsc{Erase_empty}$$

$$\frac{\Gamma = |\Delta|}{\Gamma, x{:}_{\mathit{eff}}\tau = |\Delta, x{:}_{\mathit{eff}}\tau|} \quad \textsc{Erase_eff}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce\ [\mathsf{log}]|} \quad \textsc{Erase_log}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]|} \quad \textsc{Erase_res}$$

$\boxed{\Delta' = \mathsf{affinize}(\Delta)}$

$$\frac{}{\cdot = \mathsf{affinize}(\cdot)} \quad \textsc{Aff_empty}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}_{\mathit{eff}}\tau = \mathsf{affinize}(\Delta, x{:}_{\mathit{eff}}\tau)} \quad \textsc{Aff_eff}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce\ [\mathsf{log}] = \mathsf{affinize}(\Delta, x{:}ce\ [\mathsf{log}])} \quad \textsc{Aff_log}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(0)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(0)])} \quad \textsc{Aff_res_used}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(1)])} \quad \textsc{Aff_res_avail}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(?)])} \quad \textsc{Aff_res_opt}$$

$\boxed{\mathsf{zero}(\Delta)}$

$$\frac{}{\mathsf{zero}(\cdot)} \quad \textsc{Zero_empty}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}_{\mathit{eff}}\tau)} \quad \textsc{Zero_eff}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}ce\ [\mathsf{log}])} \quad \textsc{Zero_log}$$

$$\frac{\mathsf{zero}(\Delta) \quad u \in \{0, ?\}}{\mathsf{zero}(\Delta, x{:}ce@ce'\ [\mathsf{res}(u)])} \quad \textsc{Zero_res}$$

$\boxed{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}$

$$\frac{}{\mathcal{C} = \cdot \Rightarrow \mathcal{C}} \quad \textsc{Close_empty}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \forall x{:}\tau.\, \mathcal{C}'}{\mathcal{C} = \Delta, x{:}_{\mathit{eff}}\tau \Rightarrow \mathcal{C}'} \quad \textsc{Close_eff}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow ce \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce\ [\mathsf{log}] \Rightarrow \mathcal{C}'} \quad \textsc{Close_log}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce@ce'\ [\mathsf{res}(u)] \Rightarrow \mathcal{C}'} \quad \textsc{Close_res}$$

$\boxed{\Sigma_c; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Gamma \vdash \top\ \mathsf{wf}} \quad \textsc{CtWf_top}$$

$$\frac{\Sigma_c; \Gamma \vdash \mathcal{C}_1\ \mathsf{wf} \quad \Sigma_c; \Gamma \vdash \mathcal{C}_2\ \mathsf{wf}}{\Sigma_c; \Gamma \vdash \mathcal{C}_1 \wedge \mathcal{C}_2\ \mathsf{wf}} \quad \textsc{CtWf_and}$$

$$\frac{\Sigma_c; \Gamma, x{:}_{\mathbf{S}}\tau \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_c; \Gamma \vdash \forall x{:}\tau.\, \mathcal{C}\ \mathsf{wf}} \quad \textsc{CtWf_forall}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool} \quad \Sigma_c; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_c; \Gamma \vdash ce \Rightarrow \mathcal{C}\ \mathsf{wf}} \quad \textsc{CtWf_impl}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool}}{\Sigma_c; \Gamma \vdash ce\ \mathsf{wf}} \quad \textsc{CtWf_atom}$$

$$\frac{}{\Sigma_c; \Gamma \vdash \bot\ \mathsf{wf}} \quad \textsc{CtWf_bot}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}} \cdot\ \mathsf{wf}} \quad \textsc{PfWf_nil}$$

$$\frac{\begin{array}{c}\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_c; \Gamma, x{:}_{\mathit{eff}'}\tau \vdash_{\mathit{eff}} \mathit{Pf}\ \mathsf{wf}\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}} x{:}_{\mathit{eff}'}\tau, \mathit{Pf}\ \mathsf{wf}} \quad \textsc{PfWf_comp}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool} \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}\ \mathsf{wf}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}} x{:}ce\ [\mathsf{log}], \mathit{Pf}\ \mathsf{wf}} \quad \textsc{PfWf_log}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Rightarrow \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}\ \mathsf{wf}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}} x{:}ce@ce'\ [\mathsf{res}], \mathit{Pf}\ \mathsf{wf}} \quad \textsc{PfWf_res}$$

$\boxed{\Sigma_c \vdash RF\ \mathsf{wf}}$

$$\frac{\Sigma_c; \cdot \vdash_{\mathit{eff}} \mathit{Pf}_1\ \mathsf{wf} \quad \Gamma = \mathsf{Bind}(\cdot; \mathit{Pf}_1) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}_2\ \mathsf{wf}}{\Sigma_c \vdash \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}]\ \mathsf{wf}} \quad \textsc{RFWf_rfun}$$

$\boxed{\Sigma_r \vdash f{:}RF}$

$$\frac{f{:}RF \in \Sigma_r}{\Sigma_r \vdash f{:}RF} \quad \textsc{RSLk_rfun}$$

$$\frac{f{:}\tau \to_{\mathit{eff}} \tau' \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_sig}$$

$$\frac{f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_def}$$

$\boxed{\overline{A} = \{\mathit{Pf}\}}$

$$\frac{}{[] = \{\cdot\}} \quad \textsc{PfTys_nil}$$

$$\frac{\overline{A} = \{\mathit{Pf}\}}{A{::}\overline{A} = \{x{:}_{\mathbf{0}}A, \mathit{Pf}\}} \quad \textsc{PfTys_pure}$$

$$\frac{\overline{A} = \{\mathit{Pf}\}}{\overline{A} = \{x{:}_{\mathbf{S}}\tau, \mathit{Pf}\}} \quad \textsc{PfTys_spec}$$

$$\frac{\overline{A} = \{\mathit{Pf}\}}{\overline{A} = \{x{:}ce\ [\mathsf{log}], \mathit{Pf}\}} \quad \textsc{PfTys_log}$$

$$\frac{\overline{A} = \{\mathit{Pf}\}}{\overline{A} = \{x{:}ce@ce'\ [\mathsf{res}], \mathit{Pf}\}} \quad \textsc{PfTys_res}$$

$\boxed{\Sigma_c = \mathsf{Comp}(\Sigma_r)}$

$$\frac{}{\cdot = \mathsf{Comp}(\cdot)} \quad \textsc{RSErase_empty}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, f : F = \mathsf{Comp}(\Sigma_r, f : F)} \quad \textsc{RSErase_ext}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce = \mathsf{Comp}(\Sigma_r, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce)} \quad \textsc{RSErase_fundef}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \overline{A}_1 = \{\mathit{Pf}_1\} \quad A = \mathsf{Prod}(\overline{A}_1) \quad \overline{A}_2 = \{\mathit{Pf}_2\} \quad B = \mathsf{Prod}(\overline{A}_2)}{\Sigma_c, f : A \to_{\mathit{eff}} B = \mathsf{Comp}(\Sigma_r, f : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}])} \quad \textsc{RSErase_rfun}$$

$\boxed{\vdash \Sigma_r\ \mathsf{wf}}$

$$\frac{}{\vdash \cdot \ \mathsf{wf}} \quad \textsc{RSWf_empty}$$

$$\frac{\vdash \Sigma_r \ \mathsf{wf}}{\vdash \Sigma_r, f : F \ \mathsf{wf}} \quad \textsc{RSWf_ext}$$

$$\frac{\vdash \Sigma_r \ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma, x{:}_{\mathit{eff}}\tau \vdash_{\mathit{eff}} ce \Leftarrow \tau'}{\vdash \Sigma_r, \mathsf{fun}\, f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \ \mathsf{wf}} \quad \textsc{RSWf_fundef}$$

$$\frac{\vdash \Sigma_r \ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash RF \ \mathsf{wf}}{\vdash \Sigma_r, f : RF \ \mathsf{wf}} \quad \textsc{RSWf_rfun}$$

$\boxed{\Sigma_c; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{x{:}ce\ [\mathsf{log}] \in \Delta}{\Sigma_c; \Delta \vdash x \Rightarrow ce \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_var}$$

$$\frac{\begin{array}{c} f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce' \in \Sigma_c \quad \mathit{eff} \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Delta \vdash \mathsf{unfold}\ f(ce) \Rightarrow (f\ ce) \doteq [ce/x]ce' \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_unfold}$$

$$\frac{\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathit{lpf} : ce \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_annot}$$

$$\frac{\Sigma_c; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{return}\, ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathsf{open\text{-}ret}\ \mathit{rpf} \Rightarrow ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_openret}$$

$\boxed{\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_c; \Delta \vdash \mathsf{auto} \Leftarrow ce \dashv \Delta \rightsquigarrow ce} \quad \textsc{LChk_auto}$$

$$\frac{\Sigma_c; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge (ce \Rightarrow (ce'))} \quad \textsc{LChk_sub}$$

$\boxed{\Sigma_c; \Delta \vdash x : ce@ce' \dashv \Delta'}$

$$\frac{x \neq y \quad \Sigma_c; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_c; \Delta, y{:}_{\mathit{eff}}\tau \vdash x : ce@ce' \dashv \Delta', y{:}_{\mathit{eff}}\tau} \quad \textsc{RLk_there_eff}$$

$$\frac{x \neq y \quad \Sigma_c; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_c; \Delta, y{:}ce''\ [\mathsf{log}] \vdash x : ce@ce' \dashv \Delta', y{:}ce''\ [\mathsf{log}]} \quad \textsc{RLk_there_log}$$

$$\frac{x \neq y \quad \Sigma_c; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_c; \Delta, y{:}ce''@ce'''\ [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta', y{:}ce''@ce'''\ [\mathsf{res}(u)]} \quad \textsc{RLk_there_res}$$

$$\frac{u \in \{1, ?\}}{\Sigma_c; \Delta, x{:}ce@ce'\ [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta, x{:}ce@ce'\ [\mathsf{res}(0)]} \quad \textsc{RLk_here}$$

$\boxed{\Sigma_c; \Delta \vdash \mathit{rpf} \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_c; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_c; \Delta \vdash x \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \top} \quad \textsc{RSyn_var}$$

$$\frac{\Sigma_c; \Delta \vdash \mathit{rpf} \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathit{rpf} : ce@ce' \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RSyn_annot}$$

$\boxed{\Sigma_c; \Delta \vdash \mathit{rpf} \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_c; \Delta \vdash lpf \Leftarrow ce_1 \doteq ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathsf{make\text{-}ret}\ lpf \Leftarrow \mathbf{return}\ ce_1@ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_makeret}$$

$$\frac{\Sigma_c; \Delta \vdash_{\mathit{eff}} crt \Leftarrow x{:}\mathbf{s}\tau, y{:}ce_1@x\ [\mathsf{res}], z{:}ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash \mathsf{make\text{-}take}\ crt \Leftarrow \mathbf{take}\ x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_maketake}$$

$$\frac{\Sigma_c; \Delta \vdash rpf \Rightarrow ce_1@ce_1' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash rpf \Leftarrow ce_2@ce_2' \dashv \Delta' \rightsquigarrow (\mathcal{C} \wedge (ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')} \quad \textsc{RChk_sub}$$

$\boxed{\Sigma_c; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_c; \Delta \vdash \cdot = \cdot \rightsquigarrow \top} \quad \textsc{PfEq_nil}$$

$$\frac{\Sigma_c; \Delta, x{:}_{\mathit{eff}}\tau \vdash Pf_1 = [x/y]Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash x{:}_{\mathit{eff}}\tau, Pf_1 = y{:}_{\mathit{eff}}\tau, Pf_2 \rightsquigarrow \forall x{:}\tau.\mathcal{C}} \quad \textsc{PfEq_comp}$$

$$\frac{\Sigma_c; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash x{:}ce_1\ [\mathsf{log}], Pf_1 = y{:}ce_2\ [\mathsf{log}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \mathcal{C}} \quad \textsc{PfEq_log}$$

$$\frac{\Sigma_c; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash x{:}ce_1@ce_1'\ [\mathsf{res}], Pf_1 = y{:}ce_2@ce_2'\ [\mathsf{res}], Pf_2 \rightsquigarrow (((ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')) \wedge \mathcal{C}} \quad \textsc{PfEq_res}$$

$\boxed{\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta}$

$$\frac{}{\Sigma_c \vdash \cdot \sqcap \cdot = \cdot} \quad \textsc{Merge_empty}$$

$$\frac{\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_c \vdash \Delta_0, x{:}_{\mathit{eff}}\tau \sqcap \Delta_1, x{:}_{\mathit{eff}}\tau = \Delta, x{:}_{\mathit{eff}}\tau} \quad \textsc{Merge_eff}$$

$$\frac{\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_c \vdash \Delta_0, x{:}ce\ [\mathsf{log}] \sqcap \Delta_1, x{:}ce\ [\mathsf{log}] = \Delta, x{:}ce\ [\mathsf{log}]} \quad \textsc{Merge_log}$$

$$\frac{\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta \quad u = u_0 \sqcap u_1}{\Sigma_c \vdash \Delta_0, x{:}ce@ce'\ [\mathsf{res}(u_0)] \sqcap \Delta_1, x{:}ce@ce'\ [\mathsf{res}(u_1)] = \Delta, x{:}ce@ce'\ [\mathsf{res}(u)]} \quad \textsc{Merge_res}$$

$\boxed{\Sigma_c; \Gamma \vdash q : Pf \dashv \Delta}$

$$\frac{}{\Sigma_c; \Gamma \vdash () : \cdot \dashv \cdot} \quad \textsc{RPM_empty}$$

$$\frac{\Sigma_c; \Gamma, x{:}_{\mathit{eff}}\tau \vdash q : [x/y]Pf \dashv \Delta}{\Sigma_c; \Gamma \vdash (x, q) : y{:}_{\mathit{eff}}\tau, Pf \dashv \cdot, x{:}_{\mathit{eff}}\tau, \Delta} \quad \textsc{RPM_comp}$$

$$\frac{\Sigma_c; \Gamma \vdash q : [x/y]Pf \dashv \Delta}{\Sigma_c; \Gamma \vdash (x, q) : y{:}ce\ [\mathsf{log}], Pf \dashv \cdot, x{:}ce\ [\mathsf{log}], \Delta} \quad \textsc{RPM_log}$$

$$\frac{\Sigma_c; \Gamma \vdash q : [x/y]Pf \dashv \Delta}{\Sigma_c; \Gamma \vdash (x, q) : y{:}ce@ce'\ [\mathsf{res}], Pf \dashv \cdot, x{:}ce@ce'\ [\mathsf{res}(1)], \Delta} \quad \textsc{RPM_res}$$

$\boxed{\Sigma_c; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf} \quad \Sigma_c; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} crt : Pf \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RCSyn_annot}$$

$$\frac{\Sigma_r \vdash f{:}RF \quad \mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \qquad \Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : RF \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} f\ \mathit{rsp} \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_call}$$

$$\frac{\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \qquad \Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : RF \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{prim}\ \mathit{rsp} \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_primapp}$$

$$\frac{\Sigma_c; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathsf{open\text{-}take}\ \mathit{rpf} \Rightarrow x{:}_{\mathbf{S}}\tau, y{:}ce_1@x\ [\mathsf{res}], z{:}ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_opentake}$$

$$\boxed{\Sigma_c; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\,x \to \mathit{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Rightarrow D(\tau_1, \ldots, \tau_i) \\ L : \tau\ \mathbf{in}\ D(\tau_1, \ldots, \tau_i)\ \mathbf{in}\ \Sigma_c \\ \Sigma_c; \Delta, x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\,x\ [\mathsf{log}] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow Pf \dashv \Delta', x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\,x\ [\mathsf{log}] \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\,x \to \mathit{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \forall x{:}\tau.\ ce \doteq L\,x \Rightarrow \mathcal{C}} \quad \text{CBr_branch}$$

$$\boxed{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\begin{array}{l}\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt}_1 \Rightarrow Pf' \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \\ \Gamma = |\Delta'| \quad \Sigma_c; \Gamma \vdash q : Pf' \dashv \Delta'' \\ \Sigma_c; \Delta', \Delta'' \vdash_{\mathit{eff}} \mathit{crt}_2 \Leftarrow Pf \dashv \Delta''', \Delta'''' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{len}(\Delta') = \mathsf{len}(\Delta''') \\ \mathsf{len}(\Delta'') = \mathsf{len}(\Delta'''') \\ \mathsf{zero}(\Delta'''')\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathbf{let}\ q = \mathit{crt}_1; \mathit{crt}_2 \Leftarrow Pf \dashv \Delta''' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \text{RCChk_let}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Leftarrow \mathsf{bool} \\ \Sigma_c; \Delta, x{:}ce \doteq \mathsf{true}\ [\mathsf{log}] \vdash_{\mathit{eff}'} \mathit{crt}_1 \Leftarrow Pf \dashv \Delta_1, x{:}ce \doteq \mathsf{true}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_1 \\ \Sigma_c; \Delta, x{:}ce \doteq \mathsf{false}\ [\mathsf{log}] \vdash_{\mathit{eff}'} \mathit{crt}_2 \Leftarrow Pf \dashv \Delta_2, x{:}ce \doteq \mathsf{false}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_2 \\ \Sigma_c \vdash \Delta_1 \sqcap \Delta_2 = \Delta'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathbf{if}\ [x]ce\ \mathbf{then}\ \mathit{crt}_1\ \mathbf{else}\ \mathit{crt}_2 \Leftarrow Pf \dashv \Delta' \rightsquigarrow (ce \doteq \mathsf{true} \Rightarrow \mathcal{C}_1) \wedge (ce \doteq \mathsf{false} \Rightarrow \mathcal{C}_2)} \quad \text{RCChk_if}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, \ldots, \tau_i) \\ D(a_1, \ldots, a_k) = \{L_1 : \tau_1' | \ldots | L_n : \tau_n'\} \in \Sigma_c \\ \Sigma_c; \Delta \vdash_{\mathit{eff}'} y{:}ce\ \mathsf{is}\ L_1\,x_1 \to \mathit{crt}_1 \Leftarrow Pf \dashv \Delta_1 \rightsquigarrow \mathcal{C}_1 \quad \ldots \quad \Sigma_c; \Delta \vdash_{\mathit{eff}'} y{:}ce\ \mathsf{is}\ L_n\,x_n \to \mathit{crt}_n \Leftarrow Pf \dashv \Delta_n \rightsquigarrow \mathcal{C}_n \\ \Sigma_c \vdash \sqcap(\Delta_1; \ldots; \Delta_n) = \Delta'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathbf{case}\ [y]ce\ \mathbf{of}\ \{L_1\,x_1 \to \mathit{crt}_1 | \ldots | L_n\,x_n \to \mathit{crt}_n\} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_n} \quad \text{RC}$$

$$\frac{\begin{array}{l}\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{pred}\ D(A, B) \\ \Sigma_c; \Delta \vdash_{\mathbf{0}} \mathit{crt}_1 \Leftarrow x{:}_{\mathbf{0}}A, y{:}ce@\mathit{Next}\ x\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta', x{:}_{\mathbf{0}}A, y{:}ce@\mathit{Next}\ x\ [\mathsf{res}(1)] \vdash_{\mathbf{1}} \mathit{crt}_2 \Leftarrow z{:}_{\mathbf{0}}D(A, B), y{:}ce@z\ [\mathsf{res}], \cdot \dashv \Delta', x{:}_{\mathbf{0}}A, y{:}ce@\mathit{Done}\ x\ [\mathsf{res}(1)] \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathbf{1}} \mathbf{iter}\ [ce](q = \mathit{crt}_1)\{\mathit{crt}_2\} \Leftarrow z{:}_{\mathbf{0}}B, y{:}ce@\mathit{Done}\ z\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \forall x{:}A.\ \mathcal{C}'}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathsf{exfalso} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \bot} \quad \text{RCChk_exfalso}$$

$$\frac{\begin{array}{l}\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow Pf' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta' \vdash Pf' = Pf \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCChk_sub}$$

$\boxed{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{RF} \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \multimap \mathit{Pf}\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta \rightsquigarrow \top} \quad \textsc{Spine_nil}$$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Leftarrow \tau \\ \Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathit{ce}/x]\mathit{Pf}_1 \multimap [\mathit{ce}/x]\mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathit{eff}}\tau, \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Spine_core}$$

$$\frac{\begin{array}{c}\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{lpf}, \mathit{rsp} : x{:}\mathit{ce}\ [\mathsf{log}], \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_log}$$

$$\frac{\begin{array}{c}\Sigma_c; \Delta \vdash \mathit{rpf} \Leftarrow \mathit{ce}_1 @ \mathit{ce}_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rpf}, \mathit{rsp} : x{:}\mathit{ce}_1 @ \mathit{ce}_2\ [\mathsf{res}], \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_res}$$

$\boxed{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \textsc{Tup_nil}$$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Leftarrow \tau \\ \Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathit{ce}/x]\mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathit{eff}}\tau, \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Tup_core}$$

$$\frac{\begin{array}{c}\Sigma_c; \Delta \vdash \mathit{lpf} \Leftarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{lpf}, \mathit{rsp} : x{:}\mathit{ce}\ [\mathsf{log}], \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_log}$$

$$\frac{\begin{array}{c}\Sigma_c; \Delta \vdash \mathit{rpf} \Leftarrow \mathit{ce} @ \mathit{ce}' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_c; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_c; \Delta \vdash_{\mathit{eff}} \mathit{rpf}, \mathit{rsp} : x{:}\mathit{ce} @ \mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_res}$$

$\boxed{\Sigma_c \vdash \sqcap \overline{\Delta} = \Delta}$

$$\frac{}{\Sigma_c \vdash \sqcap \Delta_0; \cdot = \Delta_0} \quad \textsc{NMerge_base}$$

$$\frac{\Sigma_c \vdash \Delta_0 \sqcap \Delta_1 = \Delta' \quad \Sigma_c \vdash \sqcap(\Delta'; \overline{\Delta}) = \Delta}{\Sigma_c \vdash \sqcap(\Delta_0; \Delta_1; \overline{\Delta}) = \Delta} \quad \textsc{NMerge_step}$$

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Definition rules:        219 good    0 bad
Definition rule clauses: 473 good    0 bad
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